Yeast Fermentation

Student worksheet | Graduated-cylinder lab (water displacement) | Biology 2e, Chapter 7

1. Plan before you mix

Names: _____ Class/date: _____

Pair/team: _____ Pooled set ID: _____ Assigned flask IDs: _____

Question: How does the available sugar affect gas accumulation in a yeast culture? Prepare your assigned one or two flasks, then pool three no-added-sugar, three glucose and three sucrose trials.

Gas enters an inverted, water-filled graduated cylinder and displaces water. The gas-water boundary provides a readable volume. This is water displacement, not collection by direct displacement of air.

Teacher-selected setup

Setting	Your assignment
Method / measurement / unit	Graduated-cylinder lab (water displacement) / Collected gas volume / mL
Flask capacity / final liquid	____ mL / ____ mL
Treatment stock / yeast suspension per flask	____ mL / ____ mL
Sugar stock / yeast stock concentration	_____ / _____
Temperature / interval / end time	____ C / ____ min / ____ min

Prediction: Compare the three treatments and explain your reasoning.

What will you change and measure? Name three things to keep the same.

Revised classroom target: 20 g yeast + 80 g sugar per 1,000 mL total water. Stock options are 20 g yeast / 500 mL water and 80 g sugar / 500 mL water; or 40 g and 160 g, respectively, per 1,000 mL water each. Water-added amounts are not exact final solution volumes.

Confirm IDs, stock labels, volumes and timing with your teacher. Wear eye protection, do not taste cultures, and never seal a culture in a rigid closed container. Follow the teacher's equipment setup instructions.

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2. Prepare and record your own flasks

Label each flask and add its assigned treatment stock. Have the yeast stock gently remixed before taking your aliquot. Add yeast last, mix consistently, promptly connect the gas collector, and start that flask's timer. Use a fresh mixture per flask.

Read the collected gas volume at eye level from the cylinder scale at the gas-water boundary. Record the initial trapped-gas volume V0 and each raw volume Vt. Record net collected volume Vt minus V0. Keep the cylinder mouth submerged and the reading position consistent.

M = missing; L = suspected leak. Record an observed zero as 0, but never replace missing data with zero.

Flask A ID: ____ Treatment: _____ Start time: ____ Temp: ____ C

Flask B ID: ____ Treatment: _____ Start time: ____ Temp: ____ C

Measurement: Collected gas volume (mL). Baseline V0: A ____ mL; B ____ mL. Net = raw Vt - V0.

Elapsed min*	A raw mL	A net mL	B raw mL	B net mL	Notes
0					
5					
10					
15					
20					
25					
30					

*Use the teacher-assigned schedule if different. Each flask has its own elapsed time. Leave B unused if you have only one flask. Do not reset or empty the collector during a trial. Stop before it reaches capacity and tell the teacher.

Describe problems or observations; keep raw readings and explain any flags.

Keep tubing unkinked and the cylinder mouth underwater. Stop before gas reaches the cylinder capacity. Disconnect the flask from the collection apparatus before cooling or dismantling to prevent water being drawn back into the culture.

Share flask IDs, readings, units, temperature and notes. Nine separate flasks are nine trials; repeated time readings from one flask are still one trial.

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3. Pool three trials of each treatment

Set ID: _____ Measurement: net collected volume (mL)

Pool matching recipes, flask sizes and equipment. Keep virtual data separate. M = missing; L = suspected leak. Record an observed zero as 0, but never replace missing data with zero. Copy net volumes, after subtracting each flask's own initial reading.

No added sugar

Elapsed min	N1	N2	N3	Mean (mL)	n
0					
5					
10					
15					
20					
25					
30					

Glucose

Elapsed min	G1	G2	G3	Mean (mL)	n
0					
5					
10					
15					
20					
25					
30					

Sucrose

Elapsed min	S1	S2	S3	Mean (mL)	n
0					
5					
10					
15					
20					
25					
30					

Mean = sum of comparable numeric readings / number of readings (n). Record actual n if readings are missing or excluded. Retain flagged readings and explain exclusions. Relabel times if the teacher assigns a different schedule.

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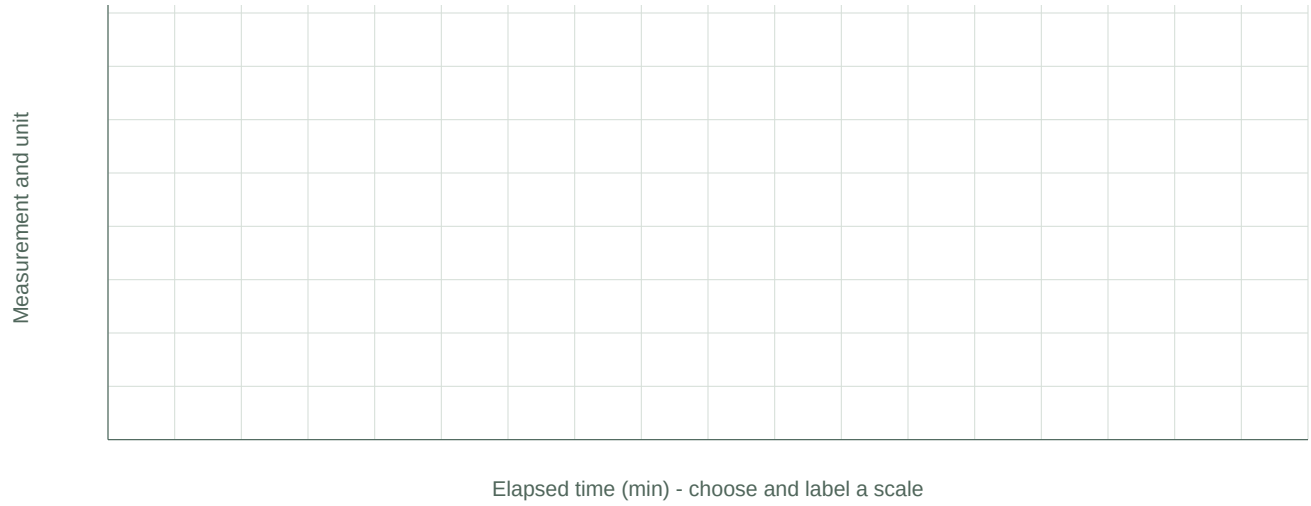
4. Compare evidence and explain

Graph net collected volume (mL) versus elapsed minutes. Label scales and units. Use different colors or symbols and a legend for the treatments. Mark missing data as gaps, not zero; label means with n when below three.



Legend: No sugar _____ Glucose _____ Sucrose _____

1. Compare treatments using numbers, units and variation among trials. Does the treatment with the highest final reading also have the fastest increase over the final interval?

2. Why include yeast with no added sugar? What would a sugar-without-yeast control help test?

3. Identify one limitation of this collection method. Why can these readings alone not measure ATP yield or prove oxygen was absent?

After the comparison: propose a follow-up variable and controls. Obtain teacher approval; keep new results separate.